Martin Andrei Urbano Santos, ECE, ECT

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Work Experience

Philippine Institute of Volcanology and Seismology (January 2025 — June 2026)

Project Technical Aide VI - Management Information System Section (COS)

- Delivered daily technical support for internal employees using Microsoft Windows, macOS, and Linux-based Desktop and Laptops, troubleshooting common issues such as software errors, printer problems, connectivity issues, and system slowdowns.
- Performed hardware troubleshooting and replacement (CPU, RAM, HDD/SSD, PSU, peripherals) to minimize downtime.
- Conducted network configuration and troubleshooting, including cabling, RJ45 crimping, LAN setup, and connectivity diagnostics to ensure connectivity.
- Installed and managed endpoint security using ESET Endpoint Security, monitoring threats and maintaining system protection.
- Deployed and configured operating systems across desktops, laptops, virtual machines, and rack servers.
- Supported system administration tasks, including server setup, SAN storage, switches, and UPS configuration.
- Maintained ICT asset inventory and ticketing systems for efficient tracking and issue resolution.
- Enhanced skills in problem-solving, network diagnostics, and technical communication, effectively translating IT issues and concepts to non-technical users.

Golden Suntec Solutions, Inc. (September 2023 — November 2023)

Software Quality Assurance Intern

- Developed comprehensive test cases and executed manual QA testing procedures to identify and validate application functionality.
- Collaborated with developers to report, prioritize, and resolve bugs to ensure high-quality software deliverables.
- Created user manuals for different applications for stakeholder's use.

Skills

Network & System Administration: Windows & Linux (RHEL 9 & Ubuntu), virtualization (VMware, Hyper-V, Proxmox), networking (TCP/IP, OSI L1–L2, switching configuration—Cisco/HP Aruba, firewalls—pfSense/Netgate), NAS management (Synology/D-Link), endpoint security (ESET), hardware support, basic web stack (Nginx, Django, Gunicorn, PHP, Laravel 8, MySQL), development (Python, Git, GitHub, Shell)

Education

Polytechnic University of the Philippines - Sta. Mesa Campus

Bachelor of Science in Electronics Engineering

Publications

Design and Development of a 6 DOF Arbiter Robotic Arm for Games of the General (Salpakan) Using Yolov8 | DOI:10.1109/IST63414.2024.10759153 | https://ieeexplore.ieee.org/document/10759153

- Trained and analysed 3 AI models (YOLOv8, YOLOv9, YOLOv10) using 300+ images that consists of 30 different classes, manually annotated and augmented using Roboflow.
- Designed a 3D model of a 6 degrees of freedom (DOF) robotic arm with Salpakan game board and stand for visualizations

AI-Read: Design and Development of a Servo-Controlled Braille Cells with OCR-Based Multimodal Braille Display for the Blind | DOI:10.1109/HNICEM64917.2024.11258782 | https://ieeexplore.ieee.org/document/11258782

- Programmed text-to-speech functionality using pyttsx3 engine resulted in 100% accuracy at 400 WPM.
- Designed a 3D model, using Fusion 360, of a working prototype with an enclosure for the microcontroller, servo-controlled braille, and camera for OCR.
- Formulated manual testing procedures and executed testing procedures to identify issues for optimization and results.
- Analyzed results and utilized different statistical treatments to interpret output and validate conclusions.

Professional Certifications

Electronics Technician Licensure Exam

Professional Regulatory Commission (PRC)

Passed: October 2025

Civil Service Pen and Paper Test - Professional

Civil Service Commission (CSC)

Passed: May 2026

Electronics Engineering Licensure Exam

Professional Regulatory Commission (PRC)

Passed: October 2026